Enhancing Operational Efficiency and Driving Business Growth at JM Marketing

A Final Report for the BDM Capstone Project

Submitted by

Name: Avnish Jajodia

Roll number: 22F3001777

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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Executive Summary –

JM Marketing, a leader in the aluminium industry, is facing long order processing times, leading to customer dissatisfaction and attrition. Moreover, the firm is encountering high holding costs and working capital requirements due to unsold inventories in the warehouse. It is also observing stagnant sales and overall business growth.
The project analysed JM Marketing's inventory and sales performance to uncover inefficiencies and growth opportunities. Data on inventory and sales across product families and brands was examined, with calculations of Net Quantity, Net Pcs and Total Movements of each SKUs. FSN analysis categorized SKUs by their movement rates, highlighting slow-moving and non-moving inventory. Sales trends were also assessed to understand consumer's brand preference and product family performance.

The analysis revealed excess inventory from underperforming brands like Local and Jindal, which led to high holding costs. Inventory analysis showed a large number of SKUs lying stagnant in the warehouse since 2022 contributing to high holding cost and overall working capital requirement of the firm. The FSN analysis provided insights on how inventory should be managed in the warehouse to reduce order processing lead times. It also identified fast-moving families and SKUs that could drive sales growth, emphasizing the need to protect these items from stockouts. On the sales side, top performers like RECT TUBE and Hindalco stood out, but overall revenue growth was stagnant, indicating inefficiencies in inventory management and market demand alignment.

JM Marketing should reduce inventory of underperforming brands and clear stagnant SKUs to optimize holding costs and free up working capital. Focus should be placed on promoting top-performing products such as RECT TUBE and Hindalco to drive revenue growth. Aligning inventories in the warehouse with FSN analysis will help in improving order processing time and help in preventing customer dissatisfaction and help in giving a competitive edge in the market.

2.Deatiled Explanation of Analysis Process –

2.1 Problem Faced by JM Marketing –

JM Marketing is encountering persistent order delays caused by prolonged processing times, leading to heightened customer dissatisfaction and increased attrition. Furthermore, the company is facing challenges with working capital being tied up in unsold inventory, resulting in higher storage costs and a negative impact on overall profitability. In addition, the firm is struggling with stagnant sales and slow business growth, compounding its operational difficulties.

The following problem statements are being analysed:

- Optimize inventory management to reduce order processing times.
- Enhance sales growth and improve working capital efficiency.

2.2 Data Collection and Cleaning –

After identifying the problems faced by the firm, specific data was collected for analysis to develop data driven solutions. The following data was gathered from the firm:

- Warehouse data
- Sales data

Upon collection, the data contained several issues and errors. These were carefully identified and cleaned before analysis, as failing to do so would have led to incorrect conclusions and ineffective solutions.

The date was not provided in the appropriate format, so it was converted to ensure efficient and accurate data analysis in the pivot table.

2.3 Data Preprocessing-

In the warehouse data, the Net Quantity (in kilograms) and Net Pieces (in units) were calculated for each Stock Keeping Unit (SKU). This represents the amount of each SKU, both by weight and in units, stored in the warehouse.

$$Net\ Quantity = Receive\ Quantity - Sale\ Quantity$$

$$Net\ Peices = Receive\ Peices - Sale\ Peices$$

The duration for which each SKU has been stored in the warehouse was also calculated by subtracting the received date from the current date and converting the result into the number of days.
This helps identify which SKUs have been in the warehouse for extended periods and assists in determining appropriate actions for such SKUs.

2.4 Descriptive Statistics:

Following the data preprocessing phase, descriptive statistics were conducted to gain a thorough understanding of the dataset. These statistics provide a valuable snapshot of the data’s key characteristics, helping to derive meaningful insights. Important measures such as central tendency (mean, median, and mode) give an idea of the data's central values, while measures of dispersion (range, variance, standard deviation) offer insight into the data's variability and distribution spread

2.5 Inventory Analysis:

An FSN (Fast, Slow, and Non-moving) analysis was conducted on the warehouse data to categorize SKUs and also families (groups of similar SKUs used to build a product) based on their movement frequency.
The total movements were calculated by summing the number of times each SKU or family was received or sold.
The formula used for calculating total movements is as follows:

$$\text{Total Movements} = \text{Count(Receive Date)} + \text{Count(Sale Date)}$$

SKUs:

The SKUs having: Total Movements >= 60: Classified as Fast.
The SKUs having: 0<Total Movements <60: Classified as Slow
The SKUs having: Total Movements < 10: Classified as Non-Moving

Families:

The families having: Total Movements >= 500: Classified as Fast.

The SKUs having: 50<Total Movements <500: Classified as Slow

The SKUs having: Total Movements < 50: Classified as Non-Moving

The classification metrics for fast, slow, and non-moving categories were established after an in-depth discussion with the founder and careful consideration of market standards.

The classification was conducted at a broader level, specifically through family-wise FSN analysis, to gain a comprehensive view of inventory trends. This analysis allowed the firm to identify the fast-moving families, enabling it to prioritize its focus on these key product groups. The classification into fast, slow, and non-moving SKUs aids in understanding the flow of inventory from the warehouse. It helped identify non-moving SKUs, which could be considered for discontinuation or disposal, thus reducing unnecessary inventory holding costs that negatively impact the firm's profitability.

This segregation also contributes to improved operational efficiency. Fast-moving SKUs can be strategically placed near the dispatch centre of the warehouse, minimizing the time workers spend searching for frequently ordered items. In contrast, slow-moving SKUs can be stored in less accessible areas of the warehouse, optimizing storage space.

By streamlining order processing, this approach leads to faster fulfilment times, ultimately improving customer satisfaction and retention. It also positions the firm to better compete in the highly competitive Chandani market of Kolkata, where timely order delivery is critical to maintaining an edge.

Unsold stocks in the warehouse since 2022 were identified by setting the receive date to 2022 and leaving the sale date blank in Excel. This approach facilitates timely elimination of outdated inventory, reducing holding costs, optimizing stock levels, and lowering the firm's overall working capital requirements.

Brand-wise unsold inventory was analysed using Net Qty to identify which brands are accumulating in the warehouse.

2.6 Sales Analysis:

The collected sales data was thoroughly analysed to identify sales trends, which provide insights into the current sales position of the firm. Brand-wise sales were calculated by grouping SKUs of similar brands, providing a comprehensive view of customer preferences for specific brands within JM Marketing's product offerings. This analysis plays a crucial role in identifying customer tastes and market demand trends. By understanding these preferences, the firm can align its strategies to cater to market demands more effectively, ensuring that it remains competitive. Additionally, this data-driven approach enables the company to focus on high-demand brands, optimize inventory, and implement targeted marketing strategies. Consequently, adapting to customer preferences not only enhances customer satisfaction but also contributes to increased sales and long-term business growth.

In addition, family-wise sales were analysed to determine which product families were contributing the most to the company's sales. This insight allows the firm to focus more on high-performing product families, enabling them to specialize and gain expertise in those areas without being distracted by managing a wide variety of product families.

This focused approach will also help reduce holding costs by allowing the company to phase out underperforming or low-demand product families, thereby increasing overall profitability. Streamlining the product portfolio in this way can enhance operational efficiency, inventory management, and financial performance.

3. Results and Findings –

3.1 Sales Trend:

The overall sales trend from August 2023 to July 2024 was analysed, revealing a sudden spike in sales during February, which was anticipated by the firm due to seasonal demand. However, outside of this peak, the overall sales remained stagnant, with a linear trend indicating a gradual decline.

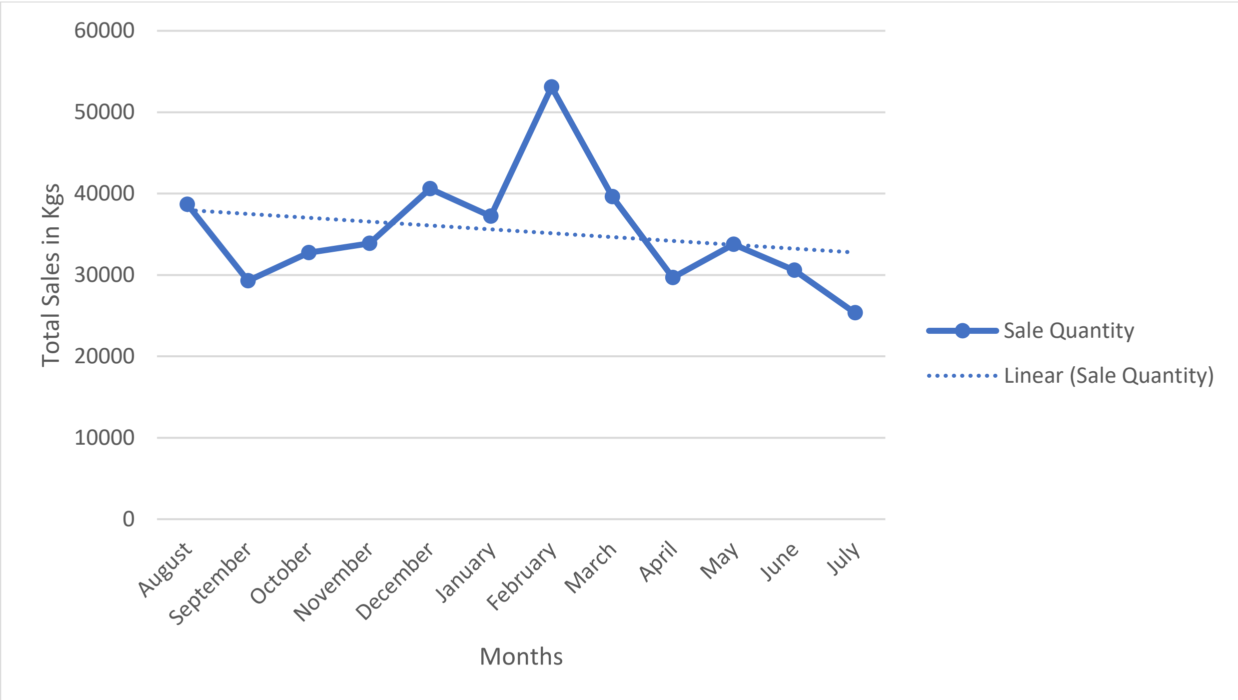

Figure 1 Sales Trend from August 2023 to July 2024.

3.2 Brand wise Sales:

The brand of a product plays a crucial role in influencing consumer purchase decisions. Therefore, analysing sales by brand helps the firm understand customer preferences and focus on high-demand brands. A 100% stacked column chart was created to compare brand-wise sales for the years 2023 and 2024.

The analysis revealed that products from Hindalco and Alom experienced overall growth in sales during this period. In contrast, Jindal's sales remained stagnant, showing no significant growth, while Local brands exhibited only marginal growth.

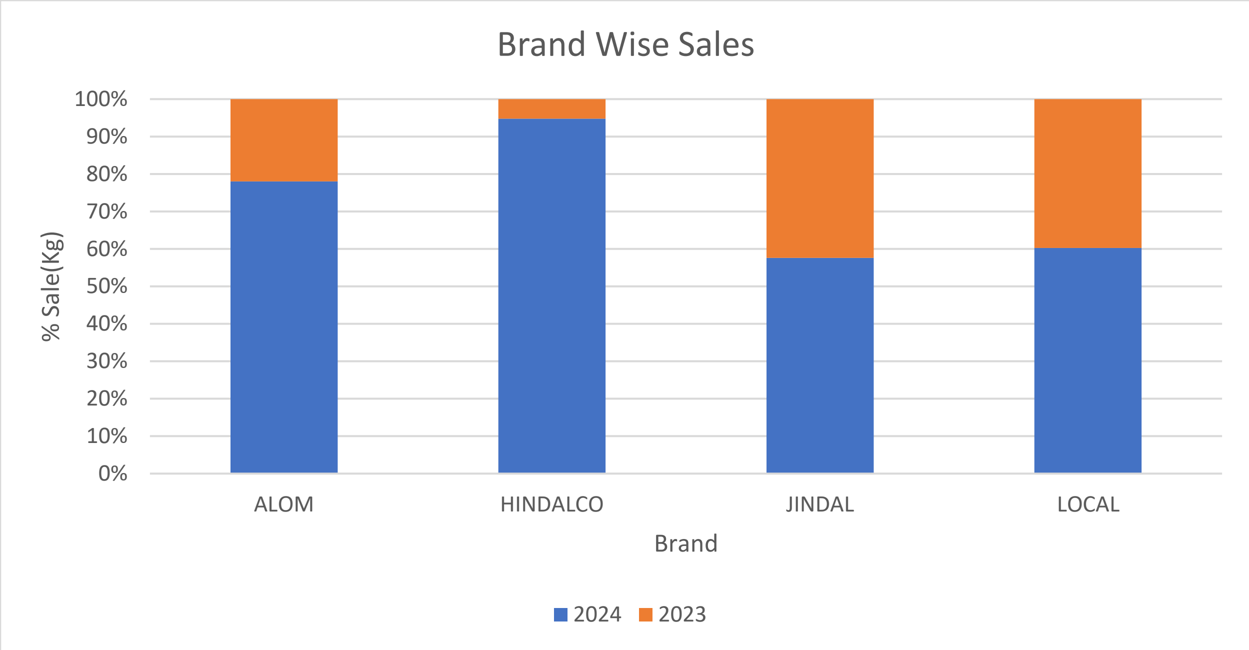

Figure 2 Comparative Brand wise Sales for year 2023 and 2024.

3.3 Brand wise Unsold Inventory in warehouse:

The pie chart clearly reveals that the Local and Jindal brands account for 97% of the warehouse inventory, while popular brands, as highlighted in the brand-wise analysis above, such as Alom and Hindalco, make up only 3% of the total inventory.

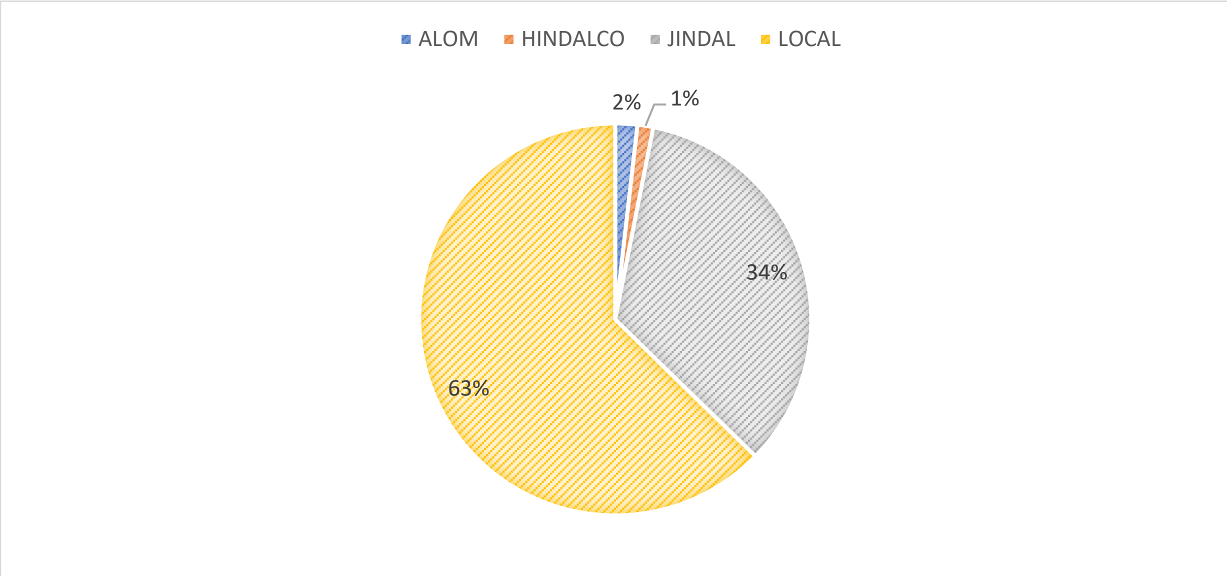

Figure 3 Brand wise unsold SKUs lying in the warehouse.

3.4 Family wise Sales Analysis:

A Pareto chart and a 100% stacked column chart were created to analyse family-wise sales for the years 2023 and 2024, facilitating a comparative analysis.

The Pareto chart reveals that the RECT TUBE family accounted for the majority of total sales, highlighting its significance in the company's revenue stream. Additionally, the families CWALL, PARTITION, SLIDING NORMAL, and SQUARE collectively contribute to approximately 80% of total sales. This concentration indicates that a few key product families are critical to the firm's performance.

In contrast, the 100% stacked column chart illustrates that while RECT TUBE experienced minimal growth in 2024, primarily driven by seasonal demand in February, the overall sales remained stagnant year-over-year. This stagnation is concerning, particularly as it indicates that the main focus area of the firm is not yielding the expected results.

The CWALL family did see reasonable sales growth, suggesting positive trends, although it does not contribute significantly to overall sales. Other families, such as PARTITION and SQUARE, exhibited nearly flat sales growth from year to year, further emphasizing the need for strategic adjustments. Notably, the SLIDING DOOR family displayed considerable growth in 2024, indicating a promising opportunity for the firm. This suggests that the company should consider focusing on this product line to capitalize on its potential and drive overall sales growth.

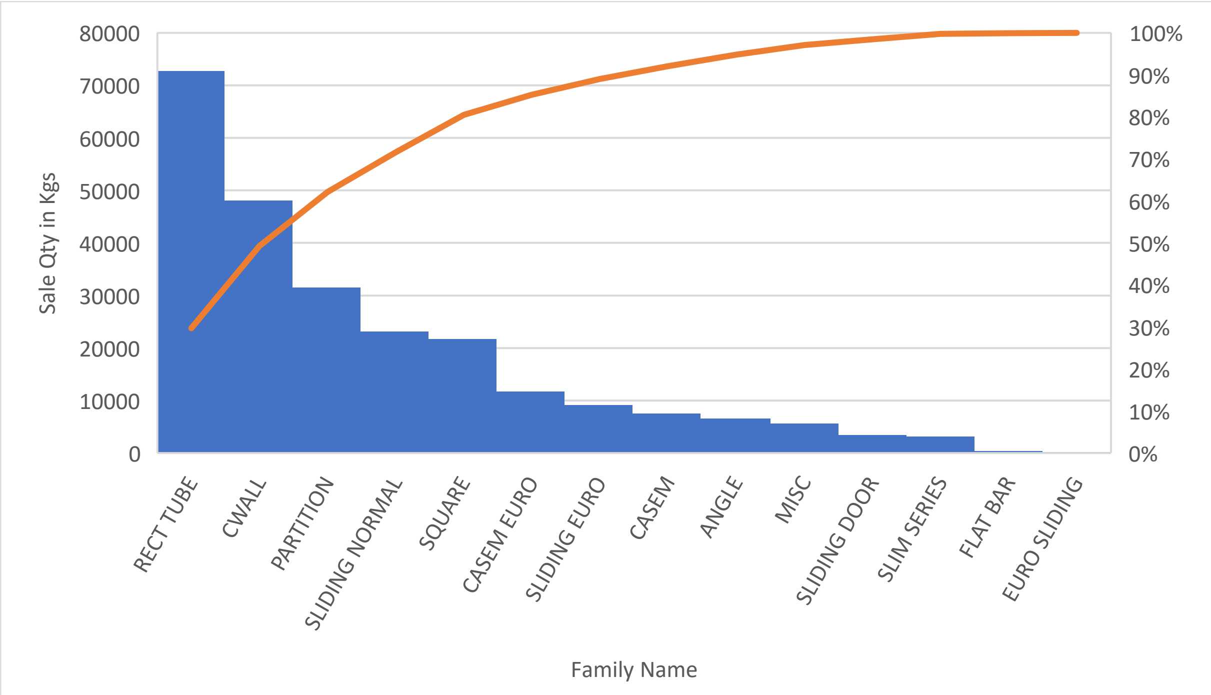

Figure 4 Pareto Chart of Family wise Sales.

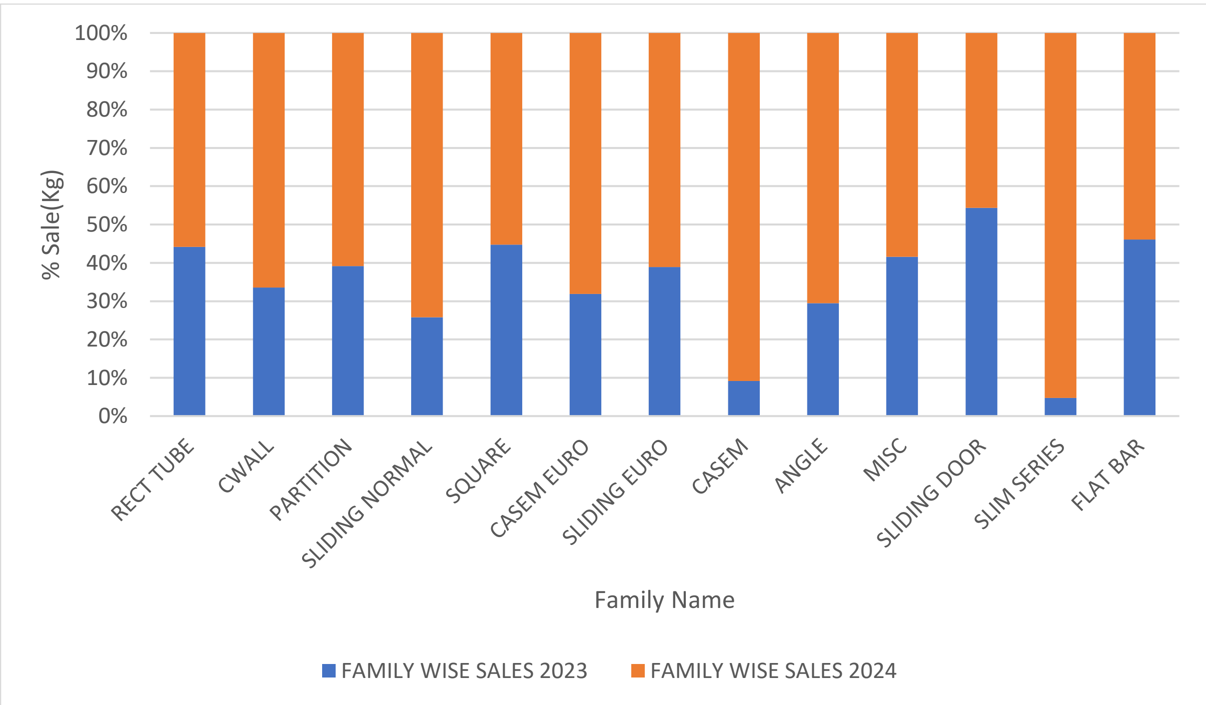

Figure 5 Comparative Family wise Sales for year 2023 and 2024.

3.5 Unsold Stock Keeping Units:

The analysis of warehouse data indicates that numerous SKUs with substantial quantities have been unsold since 2022. This situation is detrimental to the firm, as these stagnant SKUs are incurring high holding costs and causing significant blockage of working capital. As a result, this leads to decreased profitability for the firm.

SKU Lying unsold since 2022				
SKU	**FAMILY**	**Brand**	**Sum of NETT QTY**	**Sum of NETT PCS**
S.D.H JINDAL 16 FT 1.5 MM	SLIDING DOOR	JINDAL	449.35	98
RT 38*25 JINDAL 12 FT 0.5 MM	RECT TUBE	JINDAL	427.3	620
Euro Shutter W JINDAL 16 FT 1.5 MM	CASEM EURO	JINDAL	381.8	120
S.D.3TT JINDAL 16 FT 1.5 MM	SLIDING DOOR	JINDAL	357.8	45
Roof Bottom JINDAL 12 FT 1.6 MM	CWALL	JINDAL	343.2	173
Cover Plate 54*29 JINDAL 12 FT 1.5 MM	CWALL	JINDAL	224.2	132
DGU Clip Euro JINDAL 12 FT 1.1 MM	CASEM EURO	JINDAL	189.85	257
2.5" C/W Screw JINDAL 12 FT 2 MM	CWALL	JINDAL	159.75	24
S.D.4TT JINDAL 16 FT 2 MM	SLIDING DOOR	JINDAL	159.55	14
SQ 19*19 JINDAL 12 FT 1.1 MM	SQUARE	JINDAL	159.35	216
2" DV LOCAL 14 FT 2.5 MM	PARTITION	LOCAL	136.88	24
Euro 2T LOCAL 15 FT 1.5 MM	SLIDING EURO	LOCAL	118.95	32
S.D.3TB JINDAL 12 FT 2 MM	SLIDING DOOR	JINDAL	113.85	16
Euro Shutter Door LOCAL 16 FT 1.5 MM	CASEM EURO	LOCAL	111.15	26
Euro 2T LOCAL 14 FT 1.5 MM	SLIDING EURO	LOCAL	109.39	32
Euro Shutter Door LOCAL 12 FT 1.5 MM	CASEM EURO	LOCAL	103.4	32
T.S.Solid LOCAL 12 FT 3 MM	MISC	LOCAL	96.02	12
RT 75*25 JINDAL 12 FT 3 MM	RECT TUBE	JINDAL	92.25	18
DG 2.5*1.5 JINDAL 16 FT 1 MM	PARTITION	JINDAL	87.5	24
S.D.A JINDAL 16 FT 1.5 MM	SLIDING DOOR	JINDAL	80.9	20
S.D.3TB JINDAL 16 FT 2 MM	SLIDING DOOR	JINDAL	75.6	8
RT 100*45 JINDAL 12 FT 2 MM	RECT TUBE	JINDAL	72.7	12
Angle 38*25 JINDAL 12 FT 2 MM	ANGLE	JINDAL	71.8	64
S.D.I/L JINDAL 16 FT 1.5 MM	SLIDING DOOR	JINDAL	71.3	16
S.D.4TB JINDAL 12 FT 2 MM	SLIDING DOOR	JINDAL	69.75	6
Euro SGU Clip 8 MM GLS LOCAL 12 FT 1.5 M	CASEM EURO	LOCAL	69.65	96

Figure 6 Dead Sock of JM Marketing since 2022.

3.6 FSN (Fast, Slow and Non-moving) Analysis:

FSN analysis was conducted on both SKUs and families to categorize them based on their frequency of movement. This family-focused analysis enables the firm to identify which product families are in higher demand among customers. By understanding these trends, the firm can strategically concentrate its efforts on the families that generate more interest and sales.

The analysis reveals that RECT TUBE, CWALL, PARTITION, and SLIDING NORMAL are classified as fast-moving SKUs. Additionally, the Pareto analysis indicates that these families account for a significant portion of total sales, highlighting the importance of prioritizing these product lines.

Conversely, FLAT BAR and EURO SLIDING are identified as non-moving families, as their sales activity is minimal compared to other families.

FSN Analysis Family wise		
Family	**Total Movements**	**Category**
RECT TUBE	3049	FAST
CWALL	2011	FAST
PARTITION	1338	FAST
SLIDING NORMAL	933	FAST
SQUARE	870	FAST
CASEM EURO	546	FAST
SLIDING EURO	385	SLOW
MISC	297	SLOW
CASEM	292	SLOW
ANGLE	276	SLOW
SLIM SERIES	141	SLOW
SLIDING DOOR	140	SLOW
FLAT BAR	23	NON-MOVING
EURO SLIDING	11	NON-MOVING

Figure 7 Family wise FSN Analysis.

FSN analysis was conducted based on the frequency of movement of SKUs, leading to their classification into different categories. The Pie Chart indicates that the majority of SKUs are classified as slow-moving, suggesting that these items should be stored deeper within the warehouse. Conversely, approximately 28 SKUs are identified as fast-moving and should be positioned near the dispatch centre for efficient order fulfilment. Additionally, the analysis highlights the presence of non-moving SKUs, which should be discontinued and discarded to prevent unnecessary holding costs and enhance overall profitability.

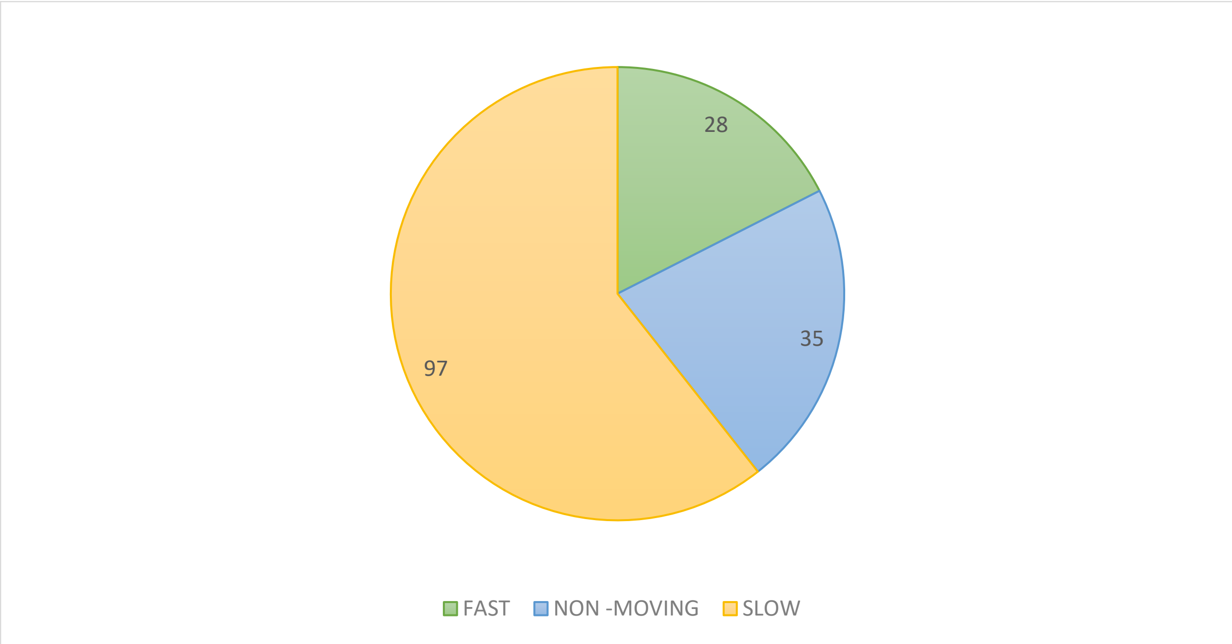

Figure 8 Pie Chart showing FSN count on SKUs.

FSN Analysis

SL.NO	SKU	Transaction	Category
1	RT 100*45 LOCAL 12 FT 1.5 MM	378	FAST
2	RT 50*25 JINDAL 12 FT 1.5 MM	206	FAST
3	RT 38*25 LOCAL 12 FT 1 MM	164	FAST
4	C/W 4" Felt LOCAL 16 FT 1.7 MM	151	FAST
5	RT 38*25 LOCAL 12 FT 0.5 MM	140	FAST
6	C/W 4" Screw LOCAL 16 FT 1.8 MM	136	FAST
7	RT 50*25 LOCAL 12 FT 1.5 MM	118	FAST
8	RT 50*25 LOCAL 12 FT 2 MM	114	FAST
9	2" C/W Screw LOCAL 16 FT 2 MM	110	FAST
10	SQ 38*38 LOCAL 12 FT 1 MM	98	FAST
11	Euro Shutter S/D LOCAL 16 FT 1.5 MI	96	FAST
12	RT 50*25 JINDAL 12 FT 2 MM	89	FAST
13	RT 75*12 LOCAL 12 FT 2 MM	88	FAST
14	RT 50*25 LOCAL 12 FT 1.3 MM	81	FAST
15	Louver Blade 3" LOCAL 16 FT 1.7 MN	78	FAST
16	C/W 131*67 JINDAL 12 FT 2 MM	74	FAST
17	C/W 81*67 LOCAL 12 FT 2 MM	71	FAST
18	C/W 4" Felt LOCAL 12 FT 1.7 MM	70	FAST
19	Cover Plate LOCAL 16 FT 1.3 MM	68	FAST
20	Pressure Plate 53 X 10 LOCAL 16 FT	68	FAST
21	RT 75*12 LOCAL 12 FT 1.5 MM	68	FAST
22	Euro Outer W LOCAL 16 FT 1.5 MM	66	FAST
23	Pressure Plate 53 X 10 LOCAL 12 FT	66	FAST
24	DGU Subframe LOCAL 16 FT 1.5 MM	65	FAST
25	SQ 50*50 LOCAL 12 FT 3 MM	65	FAST
26	C/W 81*67 LOCAL 16 FT 2 MM	64	FAST
27	SQ 19*19 LOCAL 12 FT 1.7 MM	64	FAST
28	40S Z LOCAL 16 FT 1.5 MM	60	FAST

FSN Analysis

SL.NO	SKU	Transaction	Category
29	RT 38*25 LOCAL 12 FT 2 MM	59	SLOW
30	SG 2.5*1.5 LOCAL 16 FT 1.2 MM	58	SLOW
31	Angle 50*25 LOCAL 12 FT 2 MM	57	SLOW
32	C/W 4" Screw LOCAL 12 FT 1.8 MM	57	SLOW
33	Angle 38*25 LOCAL 12 FT 1.5 MM	56	SLOW
34	SQ 50*50 JINDAL 12 FT 2 MM	55	SLOW
35	C/W 155*67 JINDAL 12 FT 3 MM	54	SLOW
36	RT 50*38 LOCAL 12 FT 1.5 MM	54	SLOW
37	I/Lock LOCAL 16 FT 1.5 MM	53	SLOW
38	SGU Subframe LOCAL 12 FT 1.6 MM	53	SLOW
39	SQ 25*25 LOCAL 12 FT 0.8 MM	53	SLOW
40	RT 100*25 LOCAL 12 FT 1.5 MM	52	SLOW
41	RT 50*25 LOCAL 12 FT 1 MM	52	SLOW
42	RT 50*12 LOCAL 12 FT 1.5 MM	51	SLOW
43	RT 50*25 JINDAL 12 FT	51	SLOW
44	SG 4*1.75 LOCAL 16 FT 3 MM	51	SLOW
45	JINDAL 12 FT	48	SLOW
46	4 1/2" DB(12MM LEG) LOCAL 16 FT 3 MM	48	SLOW
47	SQ 50*50 LOCAL 12 FT 1.5 MM	47	SLOW
48	A LOCAL 16 FT 1.5 MM	46	SLOW
49	C/W 131*67 LOCAL 12 FT 1.8 MM	46	SLOW
50	SQ 150*150 JINDAL 12 FT 3 MM	46	SLOW
51	Euro Outer Door LOCAL 16 FT 1.5 mm	45	SLOW
52	RT 100*50 LOCAL 12 FT 2 MM	45	SLOW
53	SG 4*1.75 LOCAL 16 FT 1.3 MM	45	SLOW
54	40S Outer LOCAL 16 FT 1.5 MM	44	SLOW
55	Angle 38*38 LOCAL 12 FT 2 MM	44	SLOW
56	DV 85mm LOCAL 14 FT 2 MM	44	SLOW
57	Euro Shutter S/D LOCAL 14 FT 1.5 MM	44	SLOW
58	2" C/W Screw LOCAL 12 FT 2 MM	43	SLOW
59	2" C/W Screw LOCAL 16 FT 1.8 MM	42	SLOW
60	Euro Outer W LOCAL 12 FT 1.5 MM	42	SLOW

FSN Analysis

SL.NO	SKU	Transaction	Category
126	2TB LOCAL 14 FT 1 MM	9	NON-MOVING
127	2TT JINDAL 14 FT 1.2 MM	8	NON-MOVING
128	34S Z JINDAL 16 FT 1.5 MM	8	NON-MOVING
129	Angle 12*12 JINDAL 12 FT 0.4 MM	8	NON-MOVING
130	Angle 21*21 JINDAL 12 FT 0.5 MM	8	NON-MOVING
131	Angle 38*38 JINDAL 12 FT 6 MM	8	NON-MOVING
132	ARP O/C Channel 40*40 LOCAL 12 FT 2 MM	8	NON-MOVING
133	C/W 4" Screw JINDAL 15 FT 2 MM	8	NON-MOVING
134	C/W 4" Screw LOCAL 12 FT 3 MM	7	NON-MOVING
135	DG 2*1 LOCAL 16 FT 1.5 MM	7	NON-MOVING
136	DG 2.5*1.5 LOCAL 12 FT 3 MM	7	NON-MOVING
137	DG 2.5*1.5 LOCAL 16 FT 3 MM	7	NON-MOVING
138	DGC 15*7.5 DOUBLE CHANNEL JINDAL 12 FT 0.4 MM	7	NON-MOVING
139	DGC 19*12.5 DOUBLE CHANNEL JINDAL 12 FT 0.8 MN	7	NON-MOVING
140	DGU Subframe 68.1 X 28 LOCAL 12 FT 1.5 MM	7	NON-MOVING
141	Fluted Pipe 1" LOCAL 12 FT 0.5 MM	7	NON-MOVING
142	H SEC 18mm JINDAL 16 FT 1.5 MM	7	NON-MOVING
143	Half Transome 81.16*38 JINDAL 16 FT 2 MM	7	NON-MOVING
144	I/Lock 18mm JINDAL 16 FT 1.3 MM	7	NON-MOVING
145	I/Lock 18mm LOCAL 15 FT 1.3 MM	2	NON-MOVING
146	Interlock Clip S/D REJECT LOCAL 12 FT 1.5 MM	2	NON-MOVING
147	Interlock Clip S/D REJECT LOCAL 16 FT 1.5 MM	2	NON-MOVING
148	Kitchen Profile LOCAL 12 FT 1.5 MM	2	NON-MOVING
149	Louver Blade 3" LOCAL 12 FT 1 MM	2	NON-MOVING
150	Roof Top JINDAL 12 FT 1.5 MM	2	NON-MOVING
151	RT 75*12 LOCAL 12 FT 1 MM	2	NON-MOVING
152	RT 83*38 LOCAL 12 FT 1.2 MM	2	NON-MOVING
153	RT100*20 LOCAL 12 FT 1.5 MM	2	NON-MOVING
154	RT100*25 (CB) LOCAL 12 FT 1.3 MM	1	NON-MOVING
155	S.D.2TT JINDAL 16 FT 1.5 MM	1	NON-MOVING
156	S.D.3TB JINDAL 15 FT 2 MM	1	NON-MOVING
157	S.D.3TT LOCAL 10'6" FT 2 MM	1	NON-MOVING

Figure 9 SKU wise FSN Analysis.

3.7 Trend on NET Quantity and NET Pcs:

A line chart was created to analyse the trends of Total Net Quantity (inventory weight in kilograms stored in the warehouse) and Total Net Pieces (inventory count in units) over multiple years. The chart revealed a significant increase in Total Net Quantity, indicating a substantial rise in inventory mass. However, the Total Net Pieces remained relatively stable, with minimal growth observed between 2023 and 2024.

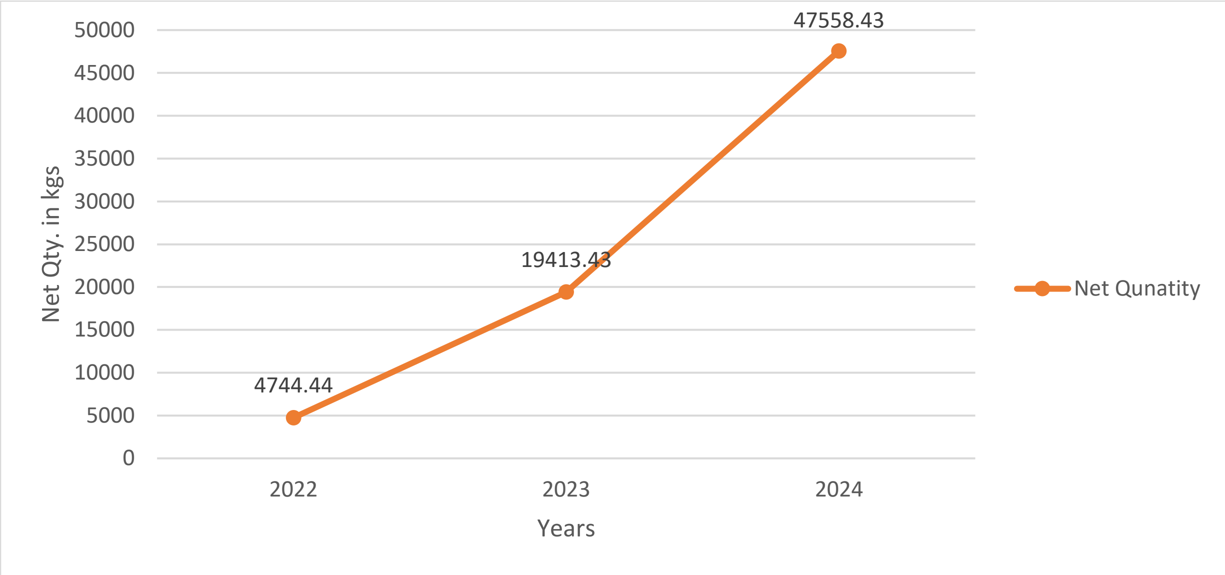

Figure 10 Trend on Net Quantity.

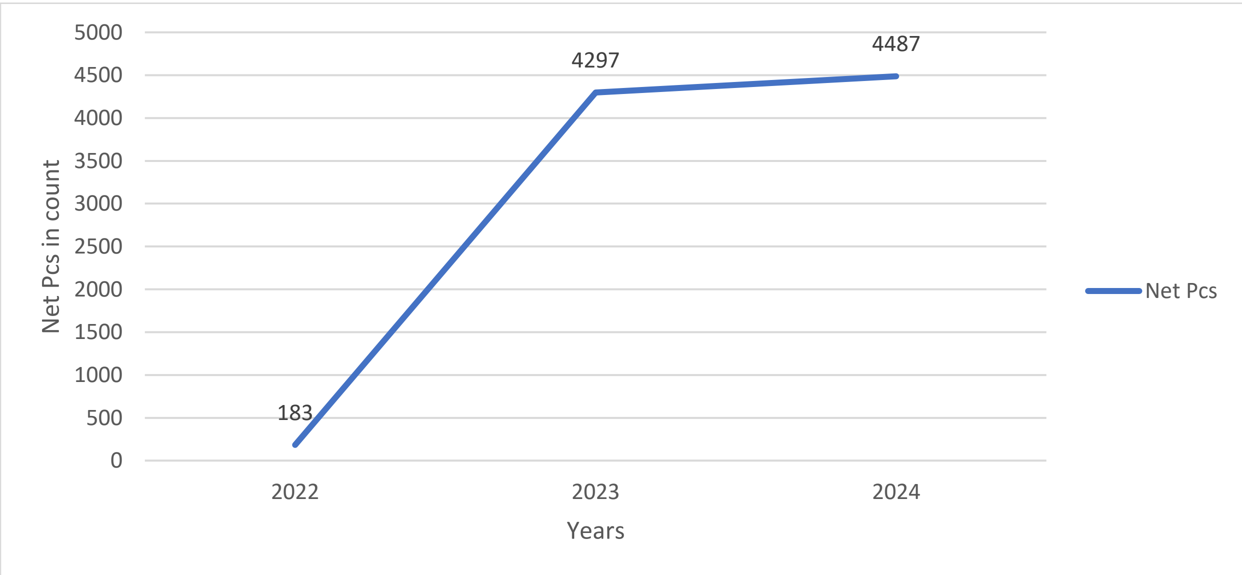

Figure 11 Trend on Net Pcs.

4. Interpretation of Results and Recommendations –

4.1 Sales Analysis:

The analysis of the overall sales trend shows a declining pattern, with the exception of a seasonal spike in February.

It was recommended that the firm implement measures such as offering discounts to stimulate sales. Additionally, distributing free samples to customers could help boost sales, enhance market presence, and improve goodwill of the firm.

4.2 Brand Analysis:

The analysis provides a brand-wise comparison of sales for 2023 and 2024 using a 100% stacked column chart. It was observed that Hindalco experienced a significant boost in sales, with 90% of its sales occurring in 2024, while the Local and Jindal brands saw only marginal growth. Alom also showed a notable increase in sales during 2024.

Additionally, the warehouse analysis revealed that SKUs for the Local and Jindal brands account for the bulk of unsold inventory, leading to increased holding costs for the company. In contrast, brands like Hindalco and Alom have much lower unsold inventory levels, indicating a stronger demand for these products. This shift in consumer buying preferences, noticeable from 2023 to 2024, is likely contributing to the disparity.

It is recommended that the firm prioritize stocking Hindalco products, as they are driving higher sales. Currently, the company predominantly focuses on Jindal and Local brands, while Hindalco represents only a small portion of the inventory. Given the changing consumer preferences, this strategy should be adjusted. By focusing more on Hindalco and Alom, rather than Jindal and Local, the firm could increase sales, minimize inventory buildup, and support overall business growth.

4.3 Family wise Sales Analysis:

he family-wise sales Pareto chart revealed that RECT TUBE, CWALL, PARTITION, SLIDING NORMAL, and SQUARE are the key product families, accounting for 80% of total sales.

It was recommended that the firm continue prioritizing these high-performing families and avoid allocating capital to underperforming families like FLAT BAR and EURO SLIDING, as they contribute minimally to sales. Investing in these low-demand families leads to unnecessary working capital blockage and higher holding costs, ultimately putting pressure on the firm's overall profitability.

Additionally, it was recommended that the firm develop strategies to further improve the sales of key performing families. The comparative analysis of 2023 and 2024 showed little to no growth in these families, despite the seasonal sales spike in February 2024. This stagnation could be a contributing factor to the firm's overall sluggish sales growth.

Addressing this issue by implementing targeted marketing, promotional activities, or product enhancements could help stimulate demand and boost revenue from these critical product families.

4.4 Net quantity and Net Pcs Analysis:

The trend in net quantity revealed that the overall total net quantity, i.e., the amount of inventory in kilograms lying in the warehouse, has seen a dramatic increase from the year 2022 to the year 2024. The increase is because the firm has increased its investments and expanded its warehouse to grow its business, which is good. However, there is a factor of unsold, outdated inventory lying in the warehouse since the year 2022, which is bad.

It was further observed that the Total Net Pieces showed little to no increase from 2023 to 2024, indicating that the rise in inventory was primarily due to weight rather than volume. While the volume remained stable, the weight of the inventory experienced a substantial increase.

It is recommended that the firm discard such SKUs as scrap to release the stuck working capital and reduce the holding costs, which have resulted in decreased profitability for the firm. The SKUs mentioned in Figure 6 have been unsold since 2022 and must be discarded to reduce the firm's burden of heavy holding costs and stuck working capital.

It is recommended to develop warehousing strategies that prioritize weight-specific inventory management rather than volume, as the volume is only marginally increasing. Focusing on weight-driven strategies will ensure better alignment with the observed trends in inventory growth.

4.5 FSN Analysis:

FSN (Fast, Slow, Non-moving) analysis was conducted on warehouse data, both at the family and SKU levels:

- Family-wise FSN:

  From the family-wise FSN analysis, it was concluded which families (groups of similar SKUs used to build a product) are fast-moving, meaning they are in higher demand compared to others. This insight allows the firm to concentrate its efforts on these high-demand families. The results aligned with our Pareto analysis, and the firm was recommended to focus solely on these SKUs to improve its market share.
- SKU-wise FSN:

  The FSN analysis was also performed at the SKU level to classify items into fast, slow, and non-moving categories, aiding in their strategic placement within the warehouse. This placement aims to reduce order processing lead times and enhance operational efficiency and customer satisfaction. It was observed that the majority of the SKUs were slow-moving, with many categorized as non-moving. It was recommended that these non-moving SKUs be discarded as scrap, as they incur unnecessary holding costs and block working capital.

  The firm was advised to arrange the SKUs according to Figure 9. Fast-moving SKUs should be positioned near the dispatch area to minimize the time workers spend searching for them, thereby enhancing order processing times. In contrast, slow-moving SKUs could be placed deeper within the warehouse.

  This analysis contributed to improving the warehouse efficiency of the firm and pinpointing the reasons for slower order processing.

It was also recommended that the firm focus more on fast-moving SKUs and families to drive business growth and increase overall sales. Additionally, it was emphasized that these items should be protected from stockouts to avoid losing potential sales opportunities.

4.6 Recommendation for Recuing Holding cost:

SKUs that have remained unsold in the warehouse since 2022 were identified, and it was recommended to the firm to discard these SKUs as scrap or sell them at a discounted price. By doing so, the company would be able to reduce the overall holding costs associated with these unsold items. The longer they remain in the warehouse, the more they contribute to higher storage expenses without generating any returns. Therefore, by either selling them at a lower price or classifying them as scrap, the company can free up warehouse space for more profitable stock, reduce the burden of holding costs, and ultimately improve its overall profitability. This strategy will not only minimize unnecessary expenses but also streamline the company's inventory management, contributing to better operational efficiency and financial performance.

4.7 Recommendation for External Competition:

- It is recommended that the firm increase its advertising budget to develop a targeted strategy aimed at standing out from the competition in the Lenin Sarani area of Chandni Market, Kolkata. A focused advertising campaign will help attract more customers, create a distinct market presence, and differentiate JM Marketing from its competitors.
- JM Marketing remains dedicated to delivering high-quality aluminium products by adhering to stringent manufacturing and transportation protocols that ensure superior standards for its customers. This approach serves as a key differentiator, reinforcing the company's Unique Selling Proposition (USP) and enhancing its competitive advantage. Therefore, it is recommended that the company continues with this strategy, as it effectively distinguishes JM Marketing from external market competition.

Data Used: BDM Project Data